

RAJEE GANESAN

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EDUCATION

Carnegie Mellon University

Ph.D. Candidate in Department of Biological Sciences
Eberly Center Future Faculty Program, Teaching Consultant Fellow

Pittsburgh, PA
April 2027

University of North Carolina, Chapel Hill

Bachelor of Science in Quantitative Biology, Minors in Data Science and Statistics; Cum Laude, Dean's List

Chapel Hill, NC
May 2022

Relevant Coursework: *Italics indicate graduate level coursework*

- **Bioinformatics:** *Bioinformatics Data Practicum, Genomics and Epigenetics of the Brain, Modeling Evolution*, Bioalgorithms, Biostatistics, Computational Sociology, Analysis of Functional Genomics Experiments
- **Computer Science:** *Practical Computing and Data Analysis*, Object-Oriented Design, Data Structures
- **Biology:** *Advanced Genetics*, Cell Biology, Molecular Biology, Evolutionary Genetics
- **Mathematics & Physics:** Vector Calculus, Decision Sciences, Statistics II, Mechanics, Electricity and Magnetism

RESEARCH EXPERIENCE

Carnegie Mellon University, Departments of Biological Science, Computational Biology

Doctoral Candidate in Andreas Pfenning Neurogenomics Lab

Pittsburgh, PA
July 2022 - Present

- *Thesis: Specialized enhancer activity associated with the convergent evolution of vocal learning and social behavior*
- Conducted computational genomics research on the molecular basis of vocal learning, integrating machine learning, comparative genomics, and epigenomics approaches. Designed and validated novel bioinformatics pipelines, developing transferable expertise in communicating complex analytical methods to diverse audiences
- Mentored undergraduate, Master's and PhD students through research projects, providing technical guidance and feedback on scientific writing and presentations
- Presented research findings regularly to interdisciplinary audiences ranging from domain experts to broader departmental communities

Terrence Furey Lab, UNC Department of Medicine

Senior Research Intern

Chapel Hill, NC
Aug 2019 - Apr 2022

- *Thesis: Allelic Imbalance and G-quadruplex analysis in ILK10 macrophages and Crohn's Disease*
- Designed and implemented bioinformatics pipelines to investigate genetic risk factors for Crohn's disease and guanine quadruplex formation
- Mentored two undergraduate students through independent pipeline adaptation projects, providing technical training and project oversight
- Completed Honors Thesis resulting in a peer-reviewed publication, developing expertise in scientific writing and translating complex findings for broader audiences

Jason Watts Lab, National Institutes of Environmental Health Sciences, National Institutes of Health

Scholars Connect Research Fellow

Durham, NC
June 2021 - April 2022

- Worked on a project to accurately identify regions forming quadruplexes, and integrated experimental results confirming a correlation between these regions and polymerase pausing/gene expression.
- Delivered communicative research presentations at biweekly meetings to technical and non-technical audiences.

Hannah Meyer Lab at Cold Spring Harbor Laboratory

Summer Research Intern

Cold Spring, NY
May 2021 - Aug 2021

- Integrated single-cell RNA-seq and ATAC-seq data to study AIRE expression in medullary thymic epithelial cells, proving that alterations in open chromatin drove autoimmune regulation in these cells, regulating self-reaction.
- Created and presented a final talk at the CSHL symposium to the department and invited guests.
- Delivered final report detailing findings and outlining next steps for full-time employees of the lab based on results.

Robert Colbert Lab at NIAMS, National Institutes of Health

Translational Pediatrics Branch Summer Research Intern

Bethesda, MD
May 2019 - Aug 2019

- Analyzed genome wide protein expression assessed by broad proteomic analysis from peripheral blood in Juvenile Dermatomyositis patients, utilizing a variety of computational softwares including R, JMP and Ingenuity Pathway Analysis.
- Organized and presented results regularly at internal meetings and delivered the final poster for NIH's Summer Poster Day.

Klaus Ley Lab at La Jolla Institute for Allergy and Immunology

Research Intern

La Jolla, CA
Jan 2019 - June 2020

- Experimentally used fluorescent and confocal microscopy in conjunction with antibody staining and image analysis software to study the biophysics of neutrophil motor function and leukocyte rolling.
- Developed understanding of research design, experimental troubleshooting, technical and protocol writing.

TEACHING EXPERIENCE

Foundations for Advancement in Sciences, NIH

BIOF555 and BIOF556: Foundations for Practical Single Cell RNA-seq Analyses & Advanced Topics

Hybrid, Bethesda, MD

Instructor, Co-instructor, Teaching Assistant

- Designed and delivered lectures, modules, and assignments across foundational and advanced single-cell RNA-seq analysis courses for a continuing education audience of researchers and clinicians at NIH. Progressed from Teaching Assistant to Co-instructor to sole Instructor across five semesters, assuming increasing ownership of course content and pedagogy
- Developed curriculum covering practical computational workflows, data interpretation, and best practices in single-cell analyses
- Facilitated regular office hours and hands-on troubleshooting sessions, guiding students through debugging computational workflows and building intuition for interpreting single-cell data

02719: Genomics and Epigenetics of the Brain | Carnegie Mellon Department of Computational Biology

Pittsburgh, PA

Guest Instructor

- Developed accessible educational tools and guided workflows enabling introductory computational biology students to run complex analyses including RERconverge and TACIT
- Mentored students through hands-on troubleshooting of bioinformatics pipelines, connecting course material and practical implementation

03220: Genetics, 03133: Neurobiology of Disease | Carnegie Mellon Department of Biology

Pittsburgh, PA

Recitation Instructor, Teaching Assistant

- Independently designed and led recitation sessions for Genetics, developing original content to reinforce and extend lecture material
- Supported both courses through exam and assignment design, grading, and individualized student instruction

Full Course History

- BIOL03220, Genetics [Recitation Instructor], Fall 2026
- BIOL02719, Genomics and Epigenetics of the Brain [Guest Instructor], Fall 2026
- BIOF555, Foundations for Practical Single Cell RNA-seq Analyses [Instructor], Fall 2026, Fall 2025
- BIOF556, Advanced Topics in Single Cell Analyses [Instructor], Fall 2026, Fall 2025
- BIOF555, Foundations for Practical Single Cell RNA-seq Analyses [Co-instructor], Spring 2025
- BIOF556, Advanced Topics in Single Cell Analyses [Co-instructor], Spring 2025
- BIOF555, Foundations for Practical Single Cell RNA-seq Analyses [Teaching Assistant, Instructor: BaDoi Phan], Fall 2024, Spring 2024
- BIOF556, Advanced Topics in Single Cell Analyses [Teaching Assistant, Instructor: BaDoi N., Ph.D.], Fall 2024, Spring 2024
- BIOL03133, Neurobiology of Disease [Teaching Assistant, Instructor: Daniel J. Brasier, Ph.D.], Spring 2026, Spring 2025, Spring 2023

MENTORSHIP RECORD

- Current Master's student, CMU (2025–present) — miRNA binding site analysis in excitatory and inhibitory neurons.
- Current undergraduate student, CMU (2025–present) — Single-cell epigenetic analysis of bipolar disorder.
- PhD student, UC San Diego. (2024–25) — Single-cell epigenetic analysis of bipolar disorder. *Best Honors Thesis, CMU School of Computer Science.*
- PhD student, Georgia Tech (2023–24) — TACIT for avian genomes for enhancer-phenotype association analysis.
- Medical student, Rutgers University. (2021–22) — Genetic colocalization in IBD.
- Medical student, UNC Chapel Hill. (2021) — Genetic susceptibility loci in Crohn's disease.

PUBLICATIONS

- Formenti G, [...] **Ganesan R**, [...] Pfenning AR, [...] Jarvis ED, et al. (*under review*) The Vertebrate Genomes Project Phase I: A global reference genome resource.
- **Ganesan R***, Li X*, Sestili H, Pfenning AR. (*in preparation*) Comparative analysis of B cell enhancer evolution associated with mammalian social behavior.
- **Ganesan R**, [...] Jarvis ED, Pfenning AR, Vernes SC, VGP Vocal Learning Group. (*in preparation*) VGP package: Shared genetic and neural features among humans and other vocal learning lineages.
- Marcus S, Davenport MH, Harter E, **Ganesan R**, Lee C, Phan BN, Terra N, Pfenning AR, Jarvis ED. (*in preparation*) Transcriptomic signatures of long-range projection class neurons for vocal control.
- **Ganesan R**, Pfenning AR. (2025) Evolution of Mammalian Regulatory Networks in the Brain. *Annual Review of Animal Biosciences*. 14. <https://doi.org/10.1146/annurev-animal-111523-102317>
- Nishiyama NC, Silverstein S, Darlington K, Kennedy Ng MM, Clough KM, Bauer M, Beasley C, Bharadwaj A, **Ganesan R**, et al. (2026) eQTL in diseased colon tissue identifies novel target genes associated with IBD. *Nature Communications*.
- **Ganesan R**, Furey T. (2022) Impact of guanine quadruplex formation on transcription and open chromatin regions in stimulated IL 10 KO macrophages. *Carolina Digital Repository*. <https://doi.org/10.17615/4vg8-aj33>
- Marki A, Buscher K, Lorenzini C, Meyer M, Saigusa R, Fan Z, Yeh YT, Hartmann N, Dan JM, Kiosses WB, Golden GJ, **Ganesan R**, et al. (2021) Elongated neutrophil-derived structures are blood-borne microparticles formed by rolling neutrophils during sepsis. *J Exp Med*. <https://doi.org/10.1084/jem.20200551>

PRESENTATIONS

- **Rajee Ganesan**, Andrew Wang, BaDoi N. Phan, Michael J. Leone, Heather Sestili Harper, Andreas Pfenning (June 2026), “Specialized enhancer activity associated with convergent evolution of vocal learning”, Evolution Conference, Cleveland, OH [Selected Talk]
- **Rajee Ganesan**, [...], Andreas Pfenning (June 2026), “Specialized enhancer activity associated with convergent evolution of vocal learning”, Genes Brain and Behavior Conference, Pittsburgh, PA [Selected Talk]
- **Rajee Ganesan**, [...], Andreas Pfenning (October 2025), “Specialized enhancer activity associated with convergent evolution of vocal learning”, CMU Computational Biology Retreat, Pittsburgh, PA [Poster]
- **Rajee Ganesan**, [...], Andreas Pfenning (October 2025), “Specialized enhancer activity associated with convergent evolution of vocal learning”, Society for Neuroscience Conference, San Diego, CA [Selected Poster]
- **Rajee Ganesan**, [...], Andreas Pfenning (October 2025), “Specialized enhancer activity associated with convergent evolution of vocal learning”, Vertebrate Genome Conference, New York City, NY [Poster]
- **Rajee Ganesan**, Matthew H. Davenport, Stephanie Marcus, Elijah Harter, BaDoi N. Phan, Andreas R. Pfenning, Erich D. Jarvis (September 2025) Transcriptomic signatures of long-range projection class neurons & specialized enhancer evolution associated with vocal learning, Department of Biological Sciences Research Club, Pittsburgh, PA [Talk]
- **Rajee Ganesan**, [...], Andreas R. Pfenning, Erich D. Jarvis (May 2025) “Transcriptomic signatures of long-range projection neurons and specialized enhancer activity for vocal control”, Center for Evolutionary Biology and Medicine, Bedford PA, [Poster]
- **Rajee Ganesan**, Ruby Redlich, Heather Harper, Amanda Kowalczyk, Andreas Pfenning (Oct 2024) “Using relative evolutionary rates and enhancer genetic variation to understand phenotypic diversity of pair bonding”, Society for Neuroscience, Chicago, IL. [Selected Poster]
- **Rajee Ganesan**, Terrence S. Furey. (April 2022) “Impact of guanine quadruplex formation on transcription and open chromatin regions in stimulated IL 10 KO macrophages”. Department of Biology Honors Thesis Symposium. [Talk]
- **Rajee Ganesan**, Terrence S. Furey. (October 2020) “Allelic Imbalance Analysis in Colonic Tissue in Crohn’s Disease”. State of North Carolina Undergraduate Research & Creativity Symposium. [Poster]

HONORS & AWARDS

- Travel Conference Award, CMU Mellon College of Sciences [Spring 2026]
- Best Poster, CMU Department of Computational Biology Retreat [Spring 2026]
- CMU Department of Biology Margaret Carver Award [Spring 2025]
- Exemplary Teaching Assistant, FAES, Bethesda, MD [Spring 2025]
- Carolina Research Scholar, UNC Chapel Hill, Chapel Hill, NC [Fall 2019 - Spring 2022]

OUTREACH & SCIENCE COMMUNICATION

UNC Tau Iota Chapter of Beta Beta Beta National Biological Honors Society | Member [AUG 2020 - MAY 2022, CHAPEL HILL, NC]
UNC Office of Undergraduate Research | Research Ambassador and Mentor [AUG 2021 - MAY 2022, CHAPEL HILL NC]
UNC Daily Tar Heel | Opinion Editor and Science Columnist [AUG 2019 - MAY 2022, CHAPEL HILL, NC]
UNC Division One Baseball | Data Analyst [DEC 2019 - DEC 2020, CHAPEL HILL, NC]
Zaniac Parkside | STEM Head Instructor, K-8 camp [JUNE 2017 - MAY 2021, CARY, NC]

SKILLS

Computational: *Languages:* Python, R, Bash, Java, SAS; *Tools:* Seurat, Scanpy, Bioconductor, GATK, JMP, Jupyter Notebooks, RMarkdown, Git/GitHub, Vim, SLURM, Microsoft Office

Research Methods: Scientific and technical writing, Research project design, Conference presentation, Antibody staining, Fluorescence and confocal microscopy, Cell and RNA isolation, Sequencing preparation

Teaching & Pedagogy: Curriculum design, Differentiated instruction, Inquiry-based learning, Inclusive pedagogy, AI literacy and integration, Student assessment, Classroom management, Hybrid and online instruction

Curriculum Development & Mentorship: Course and module design, Educational tool development, Undergraduate mentorship, One-on-one instruction, Science communication and outreach